

Data Mining Project Report

Online Retail Customer Segmentation

RFM Analysis, K-Medoid Clustering, Hierarchical Clustering, Fuzzy Logic, and Genetic Algorithm Optimization

Project Component	Description
Dataset	Online Retail II transactions from 01/12/2009 to 09/12/2010
Original Data Size	525,461 rows and 8 columns
Final Customer Matrix	4,312 customers with RFM-based features
Main Objective	Segment customers and generate actionable marketing priority scores
Main Models	K-Medoid, Hierarchical Clustering, Mamdani Fuzzy Inference System, Genetic Algorithm

1. Executive Summary

This project analyzes the Online Retail II dataset for a UK-based online retailer. The work starts with exploratory data analysis, then cleans the transaction data, builds customer-level RFM features, and applies clustering and decision-support techniques. The final result is a practical customer segmentation pipeline that assigns each customer to a business segment and recommends a marketing action based on fuzzy priority scoring.

- The raw dataset contains 525,461 transaction records across 40 countries.
- The United Kingdom dominates the dataset with about 485,852 records, approximately 92% of all transactions.
- After removing cancellations, missing Customer IDs, and non-positive quantities/prices, the clean transaction dataset contains 407,664 records.
- RFM engineering produces 4,312 unique customer profiles.
- K-Medoid clustering with $k = 4$ gives a silhouette score of 0.3745 and Davies-Bouldin index of 0.8726.
- Hierarchical clustering using Ward linkage with $k = 4$ gives a silhouette score of 0.2986 and Davies-Bouldin index of 0.9413.
- The Genetic Algorithm improves the K-Medoid silhouette score from 0.3745 to 0.6982, an 86.43% improvement over the equal-weight baseline.

2. Dataset Overview

The Online Retail II dataset contains invoice-level transactions from a UK-based non-store online retailer. Each row represents a product line in an invoice, including the invoice number, stock code, product description, quantity, invoice date, unit price, customer ID, and country.

Item	Value
Rows	525,461
Columns	8
Unique invoices	28,816
Unique stock codes	4,632
Unique product descriptions	4,681
Unique countries	40
Date range	01 Dec 2009 to 09 Dec 2010
Most common country	United Kingdom
UK transaction count	485,852

Numeric Field	Mean	Median	Min	Max	Std. Dev.
Quantity	10.34	3	-9,600	19,152	107.42
Price	4.69	2.10	-53,594.36	25,111.09	146.13
Customer ID	15,360.65	15,311	12,346	18,287	1,680.81

The negative quantity and negative price values show that the raw data includes cancellations, returns, and noisy entries. These records are removed before RFM modeling because they can distort customer value and distance-based clustering.

3. Exploratory Data Analysis and Visualizations

3.1 Transaction Volume by Country

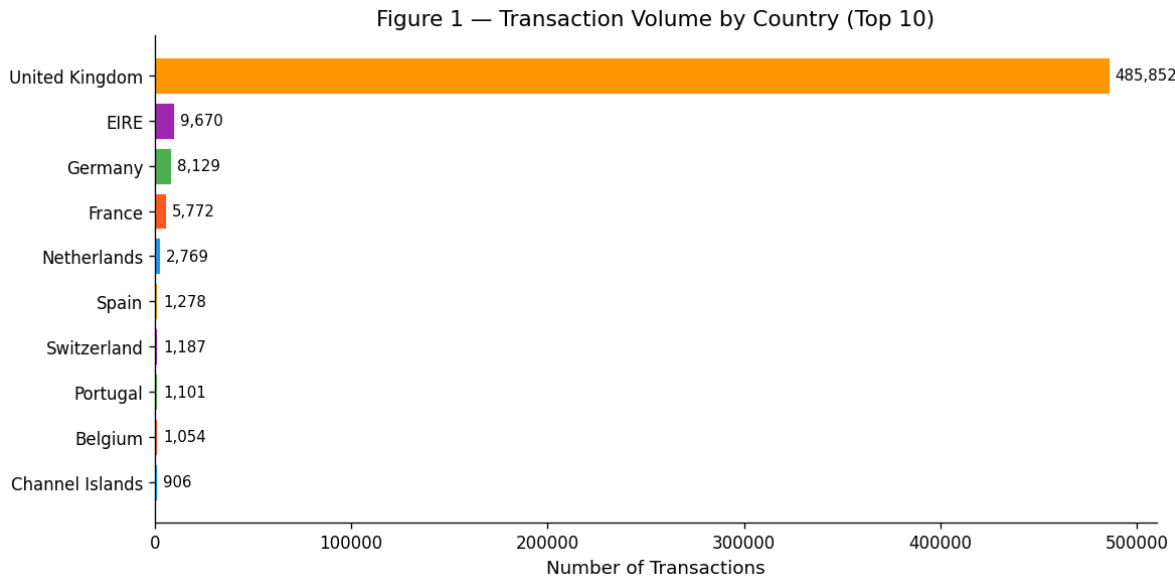


Figure 1. Top 10 countries by transaction volume.

The country distribution confirms that the project is mostly UK-centered. The United Kingdom has around 485,000 transactions, while EIRE, Germany, France, and other countries appear much less frequently. This strong geographic imbalance supports adding a UK/non-UK flag during preprocessing.

3.2 Monthly Revenue Trend

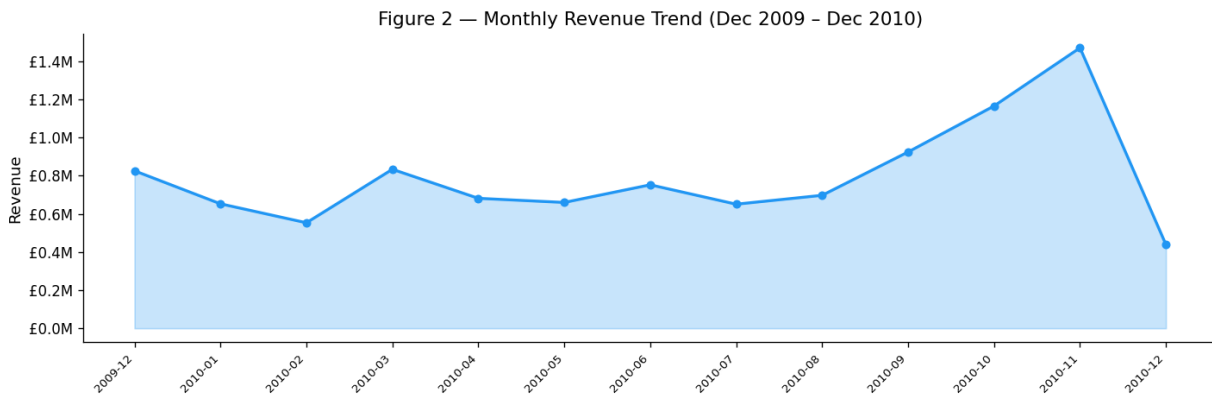


Figure 2. Monthly revenue trend after removing cancellations and invalid values.

Revenue increases strongly toward the end of 2010 and peaks in November 2010 above £1.4M. This pattern suggests seasonal demand, most likely connected to pre-Christmas shopping. Customers who purchased recently near the end of the dataset are therefore especially important in the Recency feature.

3.3 Quantity Distribution

Figure 3 — Quantity Distribution

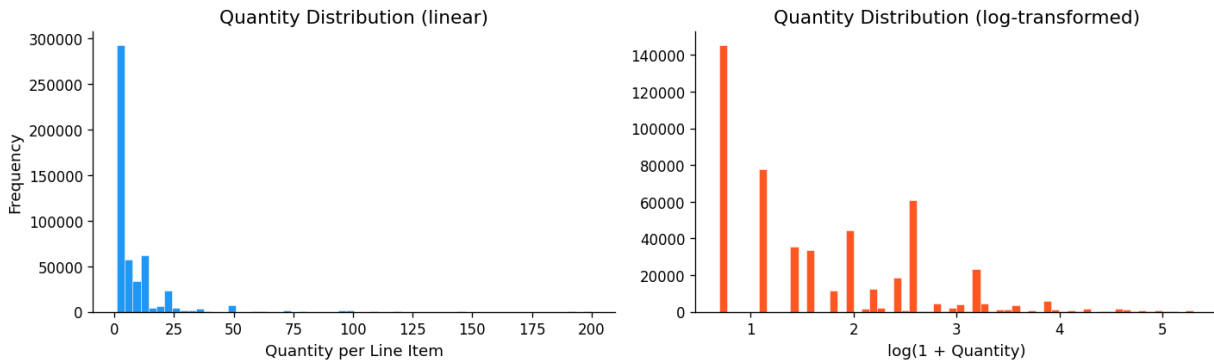


Figure 3. Quantity distribution with original and log-scale views.

Quantity is heavily right-skewed. Most transactions contain small quantities, commonly between 1 and 12 units, but some wholesale orders are much larger. This justifies log transformation and outlier handling before clustering.

3.4 Price Distribution

Figure 4 — Unit Price Distribution

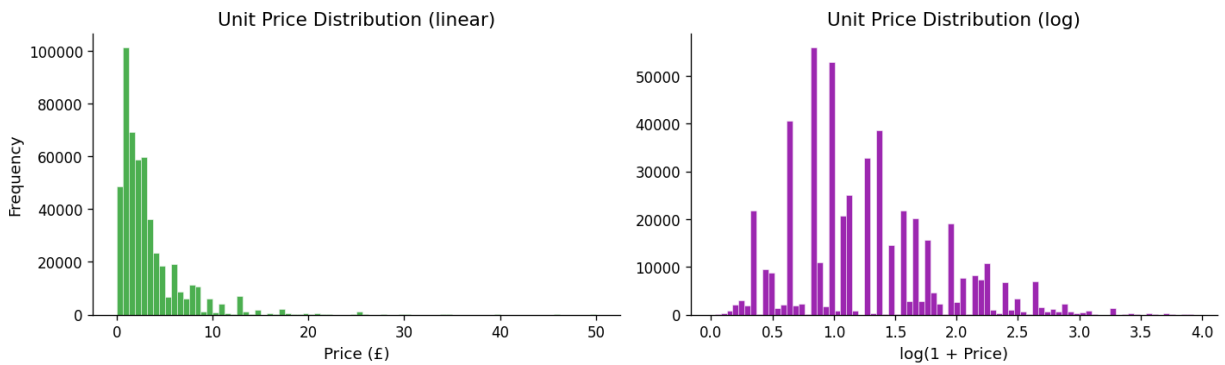


Figure 4. Unit price distribution with original and log-scale views.

Most unit prices are below £5, which fits the giftware and novelty-product nature of the retailer. The log view shows possible product tiers, including low-cost impulse items and medium-priced decorative products. Large values are treated as potential outliers.

3.5 Cancellation Rate by Country

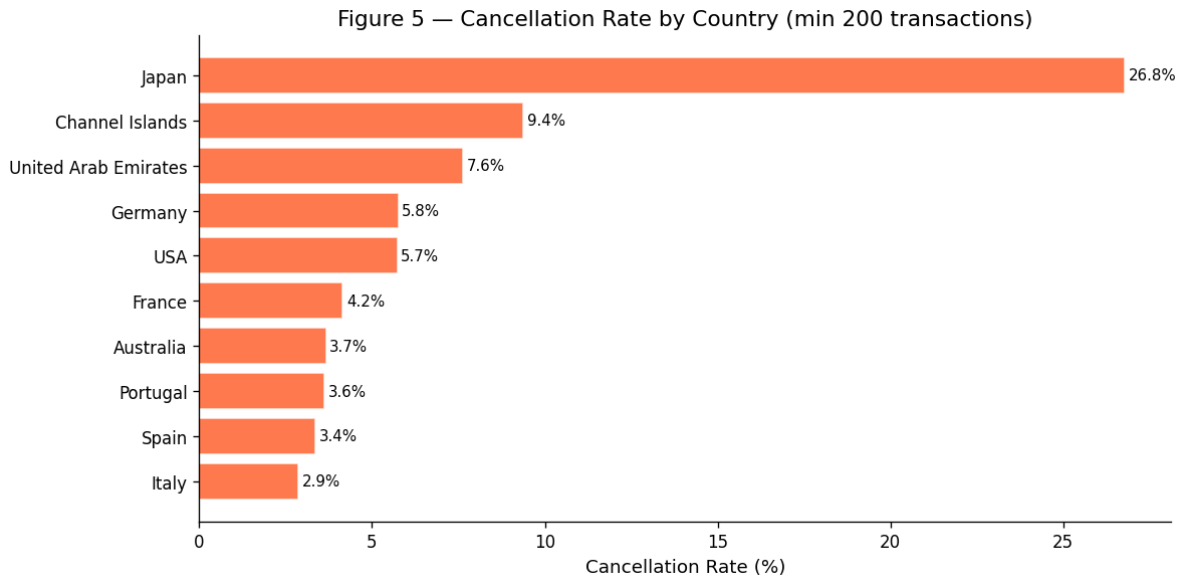


Figure 5. Cancellation rate by country for countries with enough records.

Some European markets show high cancellation rates, with some above 20%. Since cancellations can reflect returns, dissatisfaction, fulfilment issues, or data-entry adjustments, they are removed before customer-level RFM modeling.

3.6 Top Products by Revenue

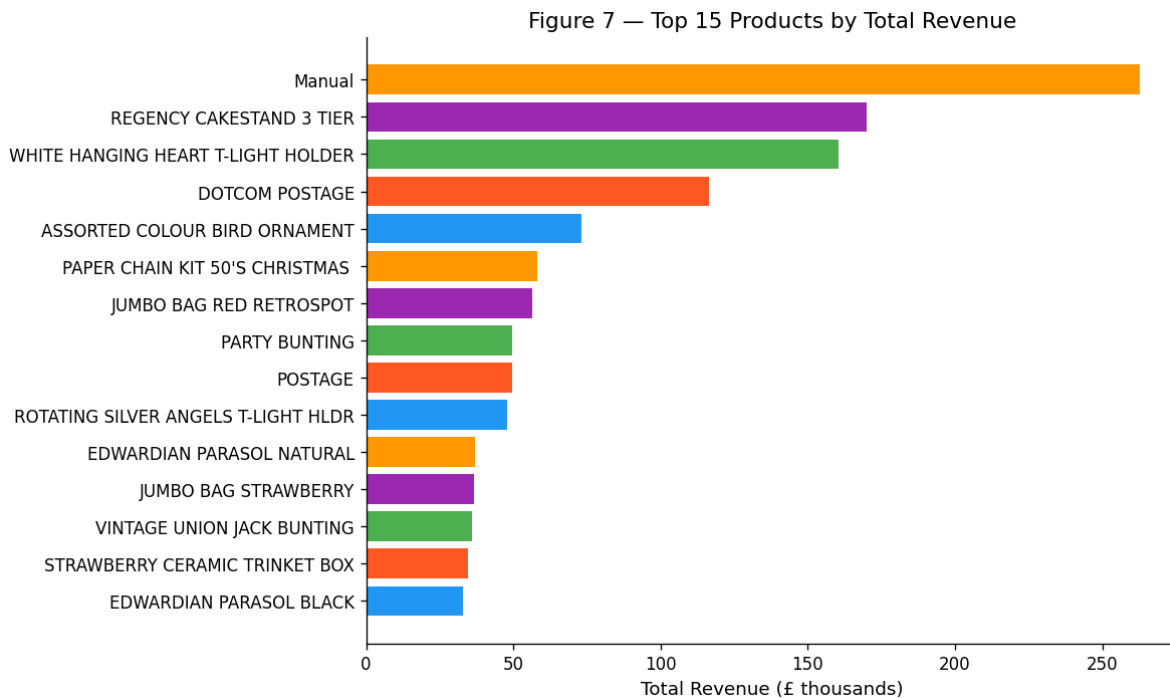


Figure 6. Top 15 products by total revenue.

The highest-revenue products include seasonal and functional items. This shows a long-tail product structure: a relatively small set of items contributes a large portion of revenue while the retailer still carries a wide product catalog.

4. Data Cleaning and Preprocessing

Step	Rows Remaining	Explanation
Raw dataset	525,461	Original imported data after adding cancellation flag.
Remove cancelled invoices	515,255	Invoices beginning with C were removed.
Drop missing Customer ID	407,695	Customer segmentation requires known customer IDs.
Remove non-positive Quantity/Price	407,664	Invalid or non-purchase records were removed.

RFM Statistic	Customer ID	Recency	Frequency	Monetary
Count	4,312	4,312	4,312	4,312
Mean	15,349.29	91.17	4.46	2,048.24
Std. Dev.	1,701.20	96.86	8.17	8,914.48
Min	12,346	1	1	2.95
25%	13,882.50	18	1	307.99
Median	15,350.50	53	2	706.02
75%	16,834.25	136	5	1,723.14
Max	18,287	374	205	349,164.35

The snapshot date used for Recency calculation is 10 Dec 2010, one day after the latest invoice date in the dataset. Frequency measures the number of unique invoices per customer, and Monetary measures total spending after cleaning.

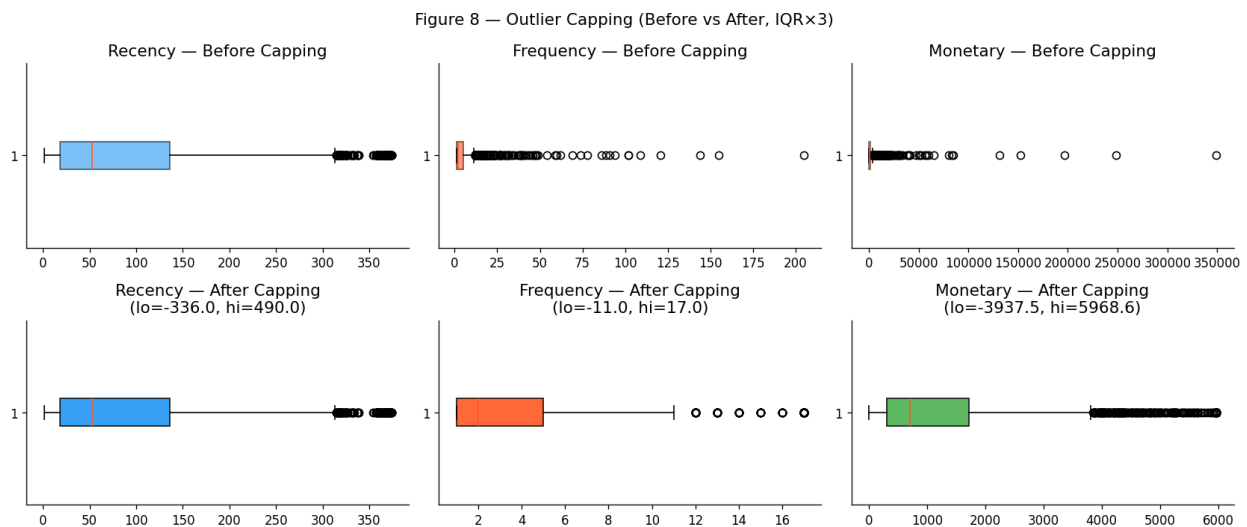


Figure 7. Before-and-after view of outlier capping for RFM variables.

After outlier capping, Frequency and Monetary are log-transformed using $\log(1+x)$. The final clustering matrix contains three numerical features: Recency, Frequency_log, and Monetary_log.

Feature	Mean	Std. Dev.	Min	Median	Max
Recency	91.17	96.86	1.00	53.00	374.00
Frequency_log	1.35	0.64	0.69	1.10	2.89
Monetary_log	6.58	1.19	1.37	6.56	8.69

PCA is used only for visualization. The first two principal components explain 93.5% of total variance, which means 2D cluster plots are meaningful for interpreting the separation between customer groups.

5. K-Medoid Clustering

K-Medoid clustering was selected because it is more robust to outliers than K-Means and because medoids are actual data points, making the resulting clusters easier to interpret. The project tests k values from 2 to 8 using inertia and silhouette score.

k	Inertia	Silhouette
2	6,857.5	0.4083
3	5,587.8	0.3332
4	4,647.1	0.3650
5	4,353.8	0.3451
6	3,946.1	0.3197
7	3,784.8	0.2582
8	3,641.8	0.2579

Figure 9 — K Selection for K-Medoid Clustering

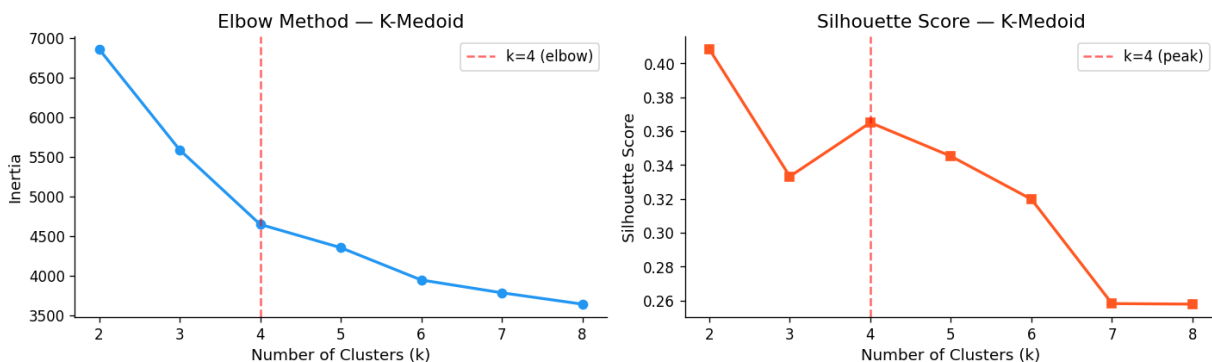


Figure 8. K-Medoid elbow and silhouette comparison.

Although k=2 has the highest silhouette score, k=4 is selected because it gives a clearer business segmentation structure and a useful elbow in inertia.

Metric	Value
Selected k	4
Final silhouette score	0.3745
Davies-Bouldin index	0.8726

Cluster	Size	Mean Recency	Mean Frequency	Mean Monetary	Business Label
0	880	257.0	1.4	£403.2	At-Risk / Lost
1	929	57.4	1.2	£279.3	At-Risk / Lost II
2	1,040	29.4	9.4	£3,533.0	Recent Buyers
3	1,463	56.8	3.2	£1,068.8	Loyal Customers

Figure 10 — K-Medoid Clusters in PCA Space

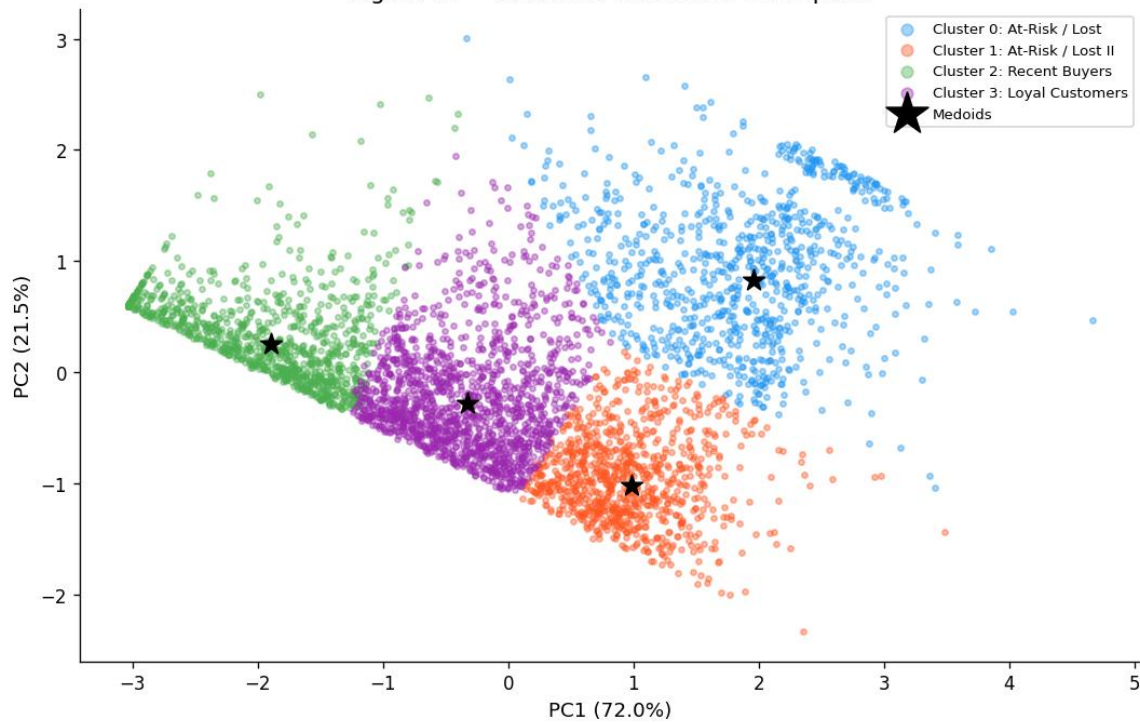


Figure 9. K-Medoid customer clusters visualized using PCA.

The K-Medoid model identifies four interpretable customer groups. The largest group is Loyal Customers with 1,463 customers. Cluster 2 has the highest average frequency and monetary value, making it the strongest commercial group even though the notebook labels it Recent Buyers.

6. Hierarchical Clustering

The project compares Ward, Complete, and Average linkage methods. Dendrograms are created using a 500-customer sample for readability, while clustering results are evaluated on the full RFM matrix.

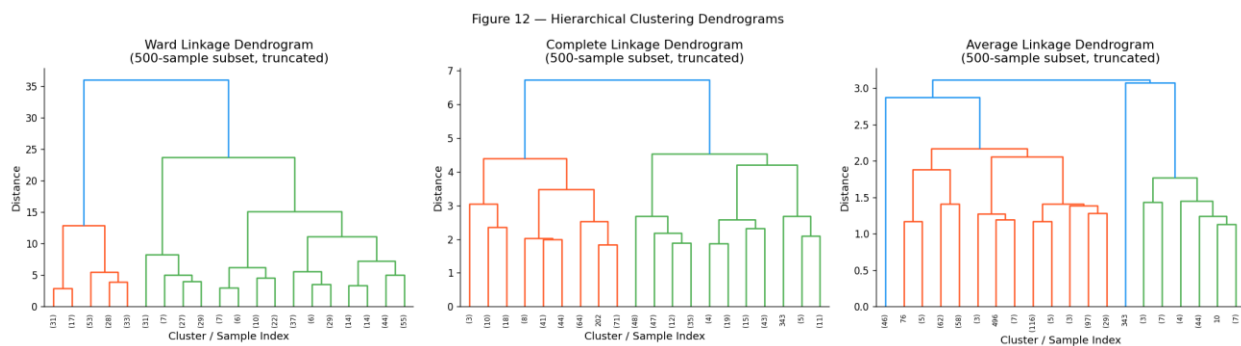


Figure 10. Truncated dendrograms for Ward, Complete, and Average linkage.

Linkage Method	Best k by Silhouette	Best Silhouette
Ward	2	0.3964
Complete	2	0.3436
Average	2	0.5660

Figure 13 — Hierarchical Clustering K Selection (3 Linkage Methods)

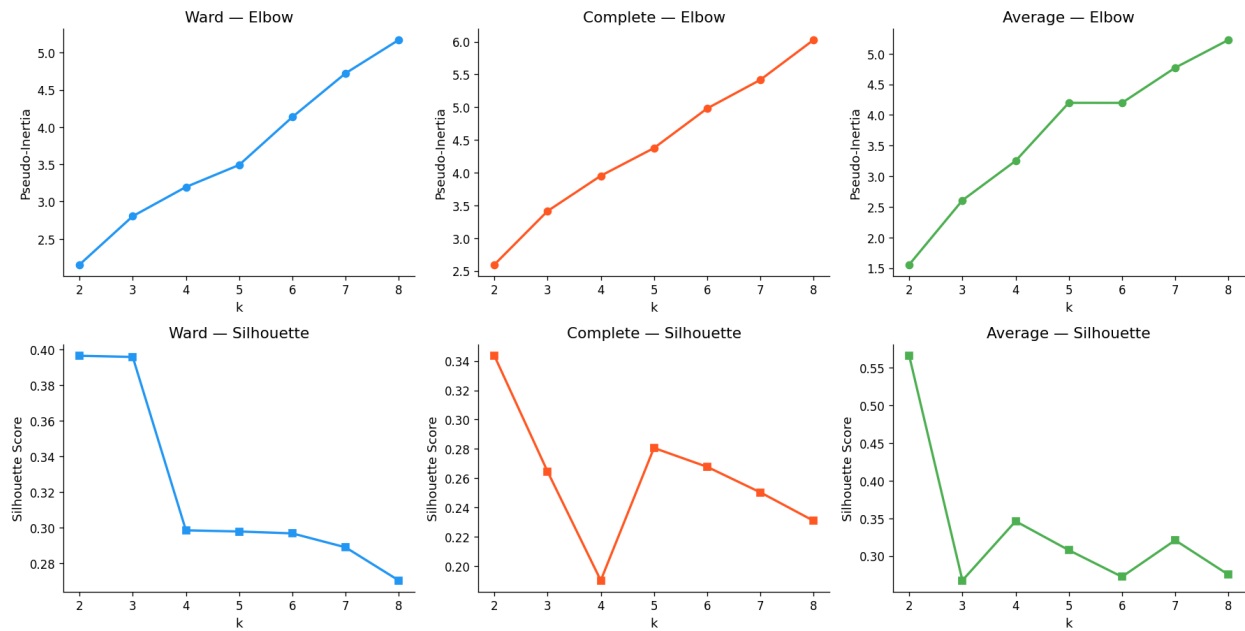


Figure 11. Hierarchical clustering elbow and silhouette comparisons.

Ward linkage with k=4 is used for final comparison with the K-Medoid four-segment solution. This keeps the business interpretation consistent across models.

Metric	Value
Selected method	Ward
Selected k	4
Silhouette score	0.2986
Davies-Bouldin index	0.9413

Hierarchical Cluster	Size
0	1,495
1	1,076
2	845
3	896

Figure 14 — Hierarchical Clusters (Ward, k=4) in PCA Space

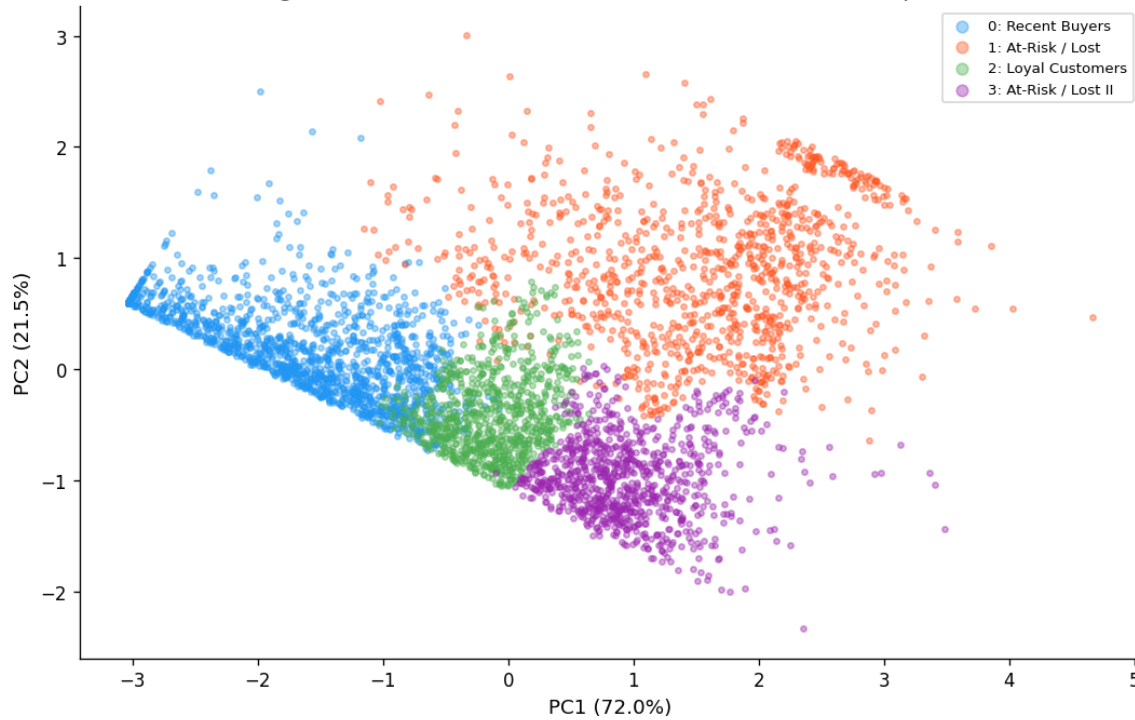


Figure 12. Hierarchical clusters visualized using PCA.

The hierarchical model supports the same general customer taxonomy as K-Medoid, although its silhouette score is lower. This suggests that K-Medoid provides a stronger final segmentation for this dataset.

7. Fuzzy Logic Inference System

The fuzzy logic system converts Recency and Monetary into a Marketing Priority Score from 0 to 100. It uses a Mamdani-style inference system with three input membership levels for each feature and four output priority levels.

Input/Output	Membership Terms
Recency	Low, Medium, High
Monetary	Low, Medium, High
Priority Score	Low, Medium, High, Very High

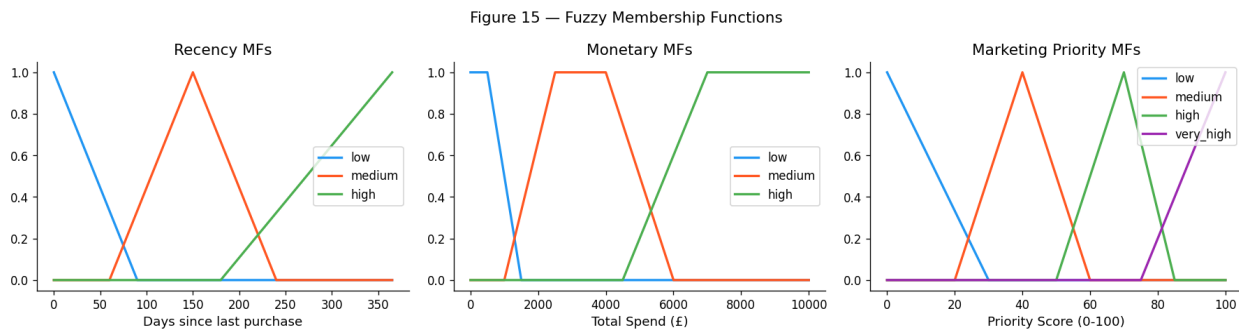


Figure 13. Fuzzy membership functions for Recency, Monetary, and Priority.

Rule	Condition	Output Priority
R1	Recency Low AND Monetary High	Very High
R2	Recency Low AND Monetary Medium	High

R3	Recency Low AND Monetary Low	Medium
R4	Recency Medium AND Monetary High	High
R5	Recency Medium AND Monetary Medium	Medium
R6	Recency Medium AND Monetary Low	Low
R7	Recency High AND Monetary High	Medium
R8	Recency High AND Monetary Medium	Low
R9	Recency High AND Monetary Low	Low

7.1 Fuzzy Validation Samples

Customer ID	Recency	Monetary	K-Med Segment	Fuzzy Priority
16858	4	£4,292.59	Recent Buyers	68.30
15194	4	£4,811.16	Recent Buyers	70.99
17524	74	£394.14	At-Risk / Lost II	30.04
16175	107	£798.85	Loyal Customers	11.54
13074	147	£1,378.74	Loyal Customers	33.36
14011	116	£457.82	Loyal Customers	11.04
18226	14	£5,968.61	Recent Buyers	89.63
17437	165	£373.17	At-Risk / Lost	10.24

Figure 16 — Fuzzy Priority Score Distribution by Customer Segment

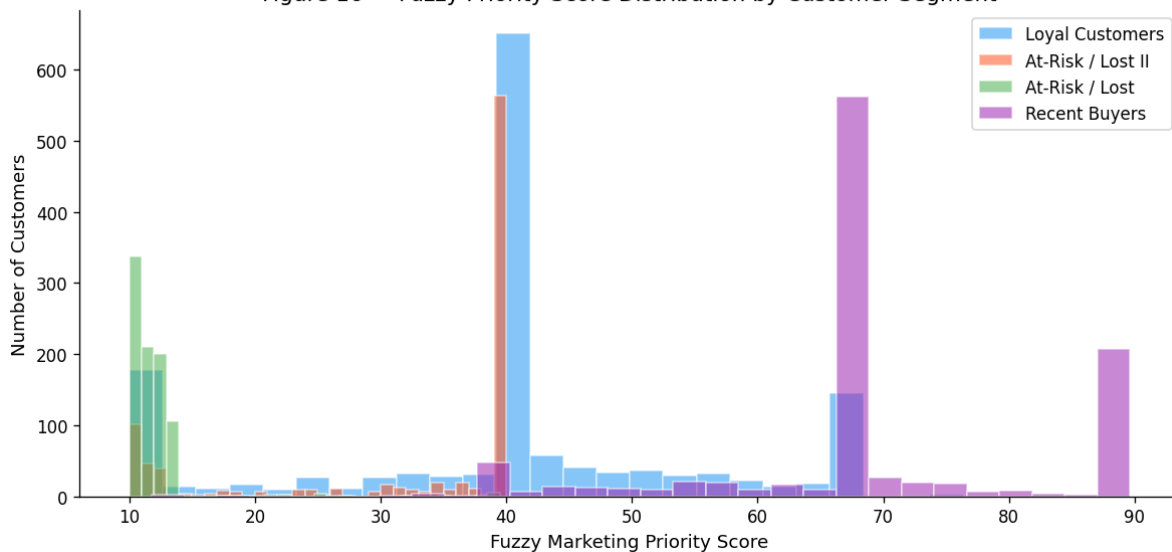


Figure 14. Fuzzy priority score distribution by customer segment.

The fuzzy system gives high priority to customers who purchased recently and spent a lot. Dormant low-value customers receive low scores, which helps the business avoid wasting campaign budget on customers with low expected return.

8. Genetic Algorithm Optimization

The Genetic Algorithm optimizes the weights applied to Recency, Frequency, and Monetary before K-Medoid clustering. The fitness function is the silhouette score, so the GA searches for the feature-weight combination that produces better cluster separation.

GA Component	Value
Chromosome	[w_Recency, w_Frequency, w_Monetary]

Population size	30 chromosomes
Generations	40
Gene range	0.1 to 3.0
Selection	Tournament selection, size 3
Crossover	Uniform crossover, 50% gene-swap probability
Mutation	Gaussian perturbation, sigma = 0.15, 20% per gene
Fitness	K-Medoid silhouette score

Generation	Best Fitness	Mean Fitness
1	0.5334	0.3982
11	0.6666	0.5798
21	0.6945	0.6407
31	0.6945	0.6537
40	0.6945	0.6379

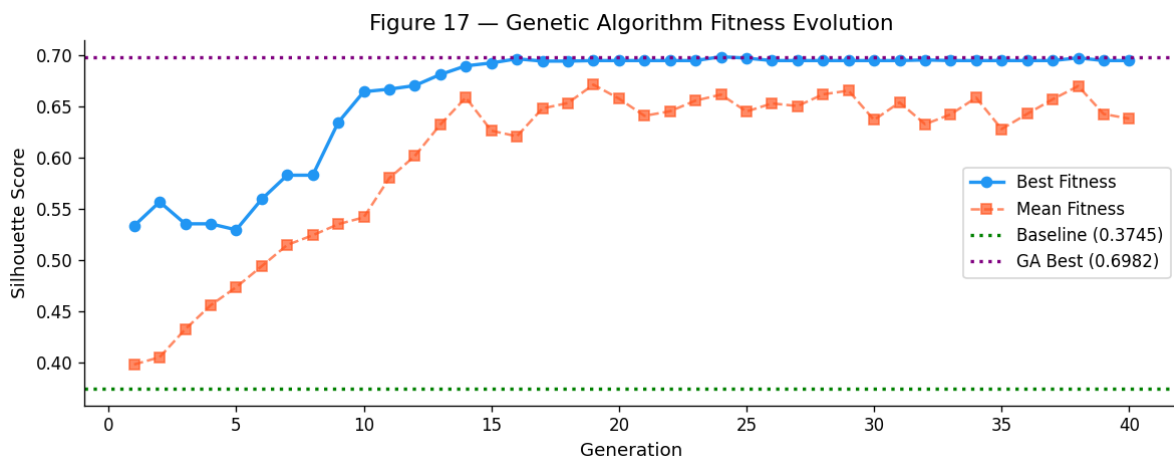


Figure 15. Genetic Algorithm fitness evolution across generations.

Result	Value
Baseline silhouette, equal weights [1,1,1]	0.3745
Best GA weights	[0.1119, 3.0000, 0.1000]
Best GA silhouette	0.6982
Improvement	+0.3237, or +86.43%
GA-weighted Davies-Bouldin	0.4365

The GA strongly improves cluster separation. The optimized weights place the largest emphasis on Frequency, while Recency and Monetary receive lower weights after scaling. This suggests that buying frequency is the strongest separator of customer behavior in the final weighted space.

9. Unified Prediction Pipeline

The final system combines preprocessing, K-Medoid clustering, Hierarchical clustering, GA-weighted clustering, and fuzzy priority scoring into one function named `predict_customer_profile()`. The function returns customer segment labels, cluster IDs, a fuzzy priority score, and a recommended marketing action.

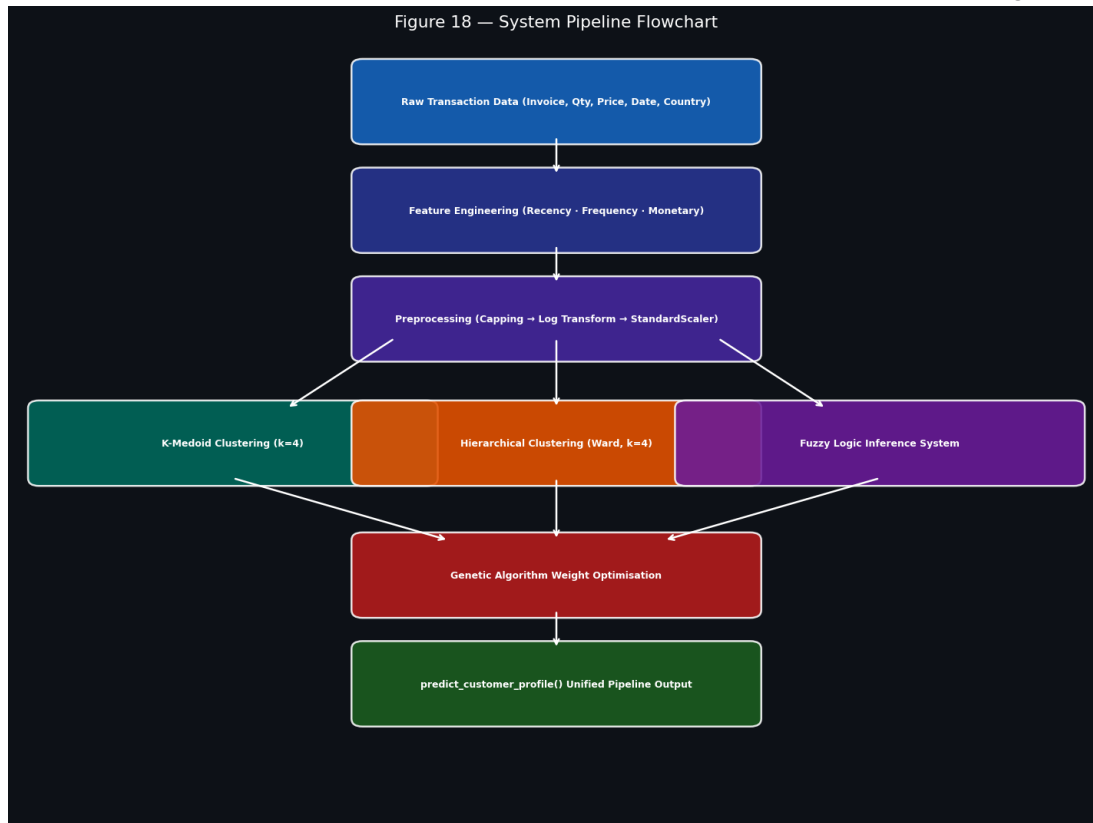


Figure 16. Unified project pipeline flowchart.

9.1 Example Customer Outputs

Customer	Recency	Frequency	Monetary	K-Medoid Segment	Fuzzy Priority	Recommended Action
14844	66 days	6	£3,512.62	Recent Buyers	61.4	Personalized re-engagement campaign
18029	278 days	1	£433.77	At-Risk / Lost	11.5	Win-back discount or archive if unresponsive
17327	45 days	2	£152.35	At-Risk / Lost II	40.0	Standard promotional newsletter
18079	127 days	1	£590.78	At-Risk / Lost II	10.5	Win-back discount or archive if unresponsive
12797	1 day	2	£552.08	Loyal Customers	40.0	Standard promotional newsletter

10. Business Interpretation

Segment / Signal	Meaning	Recommended Marketing Use
High fuzzy priority	Recent and/or high-spend customers	VIP offers, loyalty rewards, personalized campaigns
Loyal customers	Repeat buyers with moderate recency and spending	Retention offers and cross-selling
Recent buyers	Recently active customers, often with strong value potential	Nurturing and onboarding campaigns
At-risk / lost customers	Longer time since last purchase and lower value	Win-back campaigns only if expected return is reasonable

GA-optimized clusters	Better-separated customer groups after feature weighting	Use as a stronger technical segmentation baseline
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11. Conclusion

The project successfully transforms a large transaction dataset into customer-level business insights. The combination of RFM analysis and clustering produces interpretable customer segments, while the fuzzy inference system converts customer behavior into a clear priority score for marketing decisions. The Genetic Algorithm adds an optimization layer and substantially improves the clustering quality compared with equal feature weights.

- The best-performing technical result is the GA-weighted K-Medoid model with silhouette score 0.6982 and Davies-Bouldin index 0.4365.
- The most useful business output is the unified pipeline because it turns raw customer behavior into segment labels and recommended actions.
- Future improvements could include product-category segmentation, density-based outlier detection, and predictive modeling for next purchase or churn risk.

12. Technical Notes

All numeric results and figures in this report are taken from the executed outputs already saved inside the submitted notebook. One small inconsistency was found in the notebook conclusion: it mentions 4,338 customers, but the executed RFM output shows 4,312 customers. This report uses the executed output value: 4,312.